

Supplementary Material

Generation of the Bellhop3D Sample Dataset

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This supplementary document describes the complete procedure used to generate the three-dimensional (3D) underwater acoustic transmission-loss (TL) samples for deep-learning model development. The workflow combines ocean reanalysis data, bathymetric data, and Bellhop3D acoustic simulations. Each generated sample consists of environmental and source information as the model input and a corresponding 3D TL field as the prediction target.

The complete procedure is described below.

Step 1: Acquisition of oceanographic and bathymetric data

Temperature and salinity data for the target ocean region are first downloaded from the Copernicus Marine Environment Monitoring Service (CMEMS) ocean reanalysis products. The data provide three-dimensional temperature and salinity fields at approximately 50 standard vertical levels and are used to describe the spatial and temporal variability of the ocean environment.

Bathymetric data are obtained separately from the ETOPO2022 global relief dataset. The bathymetry provides the seafloor depth required to define the lower boundary of the Bellhop3D acoustic propagation environment.

For each simulation time, the required environmental information therefore consists of

$$\mathcal{E} = \{T(\lambda, \phi, z), S(\lambda, \phi, z), D(\lambda, \phi)\}, \quad (1)$$

where T and S denote temperature and salinity, respectively, D denotes water depth, and (λ, ϕ, z) represent longitude, latitude, and depth.

Step 2: Conversion of temperature and salinity to sound speed

The CMEMS temperature and salinity fields are converted into sound-speed fields using the Chen–Millero sound-speed equation. At each horizontal grid point and vertical level, the sound speed is calculated as

$$c(S, T, P) = C_w(T, P) + A(T, P)S + B(T, P)S^{3/2} + D(T, P)S^2, \quad (2)$$

where S , T , and P denote salinity, temperature, and pressure, respectively. The terms C_w , A , B , and D are polynomial functions of temperature and pressure.

After this conversion, each oceanographic time step is represented by a three-dimensional sound-speed field,

$$c = c(\lambda, \phi, z). \quad (3)$$

This field is subsequently used to extract the sound-speed profile corresponding to each source position.

Step 3: Definition of source locations and extraction of vertical sound-speed profiles

A set of candidate acoustic source locations is predefined within the target region. Each source is specified by its longitude and latitude,

$$\mathbf{s} = (\lambda_s, \phi_s). \quad (4)$$

In the experiments reported in the main manuscript, the study region is discretized into an 11×11 grid of candidate source locations.

For each source position, the vertical sound-speed profile (SSP) is extracted from the 3D sound-speed field at the corresponding horizontal coordinate. The profile retains the approximately 50 standard CMEMS depth levels,

$$\mathbf{c}_s = [c_1, c_2, \dots, c_{50}]^T, \quad (5)$$

where c_k denotes the sound speed at the k th standard depth.

At relatively shallow source locations, some deeper CMEMS levels may lie below the local seafloor. These levels are generally represented by missing values in the original ocean reanalysis product. To maintain a fixed 50-dimensional SSP representation for all samples, the sound speed at these below-bottom levels is manually assigned a value of

$$c_{\text{bottom}} = 1800 \text{ m/s}, \quad (6)$$

which is consistent with the seabed sound-speed parameter adopted in the Bellhop3D simulations.

The longitude, latitude, and 50-dimensional SSP extracted in this step are stored and later used as the environmental input of the corresponding deep-learning sample.

Step 4: Extraction of local bathymetry around each source

For each source position, the surrounding seafloor topography is extracted from the ETOPO2022 dataset.

A local bathymetric domain extending approximately 60 km around the source is selected. This domain is chosen to be larger than the maximum 50 km receiver range used in the acoustic calculations, ensuring that the entire acoustic propagation domain is covered by the extracted bathymetry.

The local bathymetry can be represented as

$$D_s = D(\lambda, \phi), \quad (\lambda, \phi) \in \Omega_s, \quad (7)$$

where Ω_s denotes the local horizontal domain centered on source location $\mathbf{s}$.

The extracted bathymetric data are then transformed into the coordinate and format requirements of Bellhop3D and used to define the spatially varying seafloor boundary.

Step 5: Construction of the Bellhop3D environmental files

The SSP obtained in Step 3 and the local bathymetry obtained in Step 4 are embedded into the Bellhop3D environmental input files.

For each source and environmental state, a complete Bellhop3D simulation configuration is generated. The configuration includes the source parameters, sound-speed profile, seabed properties, bathymetric information, receiver geometry, and numerical ray-tracing parameters.

The principal parameters used in this study are summarized in Table 1.

Table 1: Principal Bellhop3D parameters used for sample generation.

Parameter	Setting
Source frequency	1000 Hz
Source depth	50 m
Maximum receiver range	50 km
Horizontal range resolution	200 m
Azimuthal coverage	0°–360°
Azimuthal resolution	10°
Receiver depths, Region 1	1000–4000 m
Receiver-depth interval, Region 1	1000 m
Receiver depths, Region 2	500–2000 m
Receiver-depth interval, Region 2	500 m
Seabed density	1.8 g/cm ³
Seabed sound speed	1800 m/s
Seabed attenuation	0.8 dB/λ
Ray beams	1000 per azimuthal profile
Acoustic-field summation	Incoherent

With these settings, the receiver grid contains four depth levels, 36 azimuthal directions, and 250 range points. Consequently, the acoustic field associated with each simulation has the dimension

$$N_z \times N_\theta \times N_r = 4 \times 36 \times 250. \quad (8)$$

The Bellhop3D parameters used here are identical to those reported in the main manuscript.

Step 6: Execution of Bellhop3D

After the environmental files have been generated, they are supplied to Bellhop3D for forward acoustic simulation.

For a given source position and oceanographic state, Bellhop3D performs three-dimensional ray tracing through the specified sound-speed and bathymetric environment. The contributions of the propagated rays are combined to obtain the acoustic pressure field at all predefined receiver locations.

The calculated acoustic field is written by Bellhop3D to a `.shd` file. Thus, one Bellhop3D run can be represented schematically as

$$\{\text{source, SSP, bathymetry, simulation parameters}\} \longrightarrow \text{Bellhop3D} \longrightarrow \text{.shd}. \quad (9)$$

Each source location and each oceanographic state requires an independent Bellhop3D simulation.

Step 7: Extraction of acoustic pressure and calculation of TL

The Bellhop3D `.shd` file is subsequently read to extract the complex acoustic pressure at every receiver-grid point.

Let

$$p(z, \theta, r) \quad (10)$$

denote the complex acoustic pressure at receiver depth z , azimuth θ , and range r . The corresponding transmission loss is calculated as

$$\text{TL}(z, \theta, r) = -20 \log_{10} \left[\frac{|p(z, \theta, r)|}{|p_0|} \right], \quad (11)$$

where p_0 is the reference acoustic pressure.

The calculated TL values are organized according to receiver depth, azimuth, and range to form a three-dimensional tensor,

$$\mathbf{Y} = \text{TL}(z, \theta, r) \in \mathbb{R}^{4 \times 36 \times 250}. \quad (12)$$

This tensor represents the Bellhop3D reference acoustic field associated with the current source position and environmental state.

Step 8: Construction of a single supervised sample

For each Bellhop3D calculation, the source information obtained in Step 3 is used as the input of one deep-learning sample.

The input vector is formed by concatenating the source longitude, source latitude, and the 50-dimensional SSP,

$$\mathbf{q} = [\lambda_s, \phi_s, c_1, c_2, \dots, c_{50}]^T \in \mathbb{R}^{52}. \quad (13)$$

The corresponding 3D TL field obtained in Step 7 is used as the target,

$$\mathbf{Y} \in \mathbb{R}^{4 \times 36 \times 250}. \quad (14)$$

Therefore, a single supervised sample is expressed as

$$\mathcal{D}_n = (\mathbf{q}_n, \mathbf{Y}_n). \quad (15)$$

Here, $\mathbf{q}_n$ contains the source position and time-dependent SSP, while $\mathbf{Y}_n$ contains the corresponding Bellhop3D-simulated 3D TL field.

The bathymetric information is not directly included in the 52-dimensional sample vector. Instead, it influences the reference TL field through the Bellhop3D simulation and is subsequently incorporated into the geo-steady-field prior used by CMSM-Net.

Step 9: Large-scale sample generation and dataset partitioning

Steps 3–8 are repeated for all source locations and all available oceanographic states.

For each year and month, the SSP associated with each predefined source location is extracted, the corresponding local bathymetry is prepared, and an independent Bellhop3D simulation is performed. This procedure generates a large collection of paired samples,

$$\mathcal{D} = \{(\mathbf{q}_n, \mathbf{Y}_n)\}_{n=1}^N. \quad (16)$$

For the two study regions considered in this work, the data cover 13 years, 12 months per year, and 11×11 source locations. Therefore, the total number of Bellhop3D samples is

$$N = 2 \times 13 \times 12 \times 11 \times 11 = 37\,752. \quad (17)$$

The generated data are finally divided chronologically into training, validation, and test sets:

$$\begin{aligned} \text{Training set:} & \quad 2012\text{--}2022, \\ \text{Validation set:} & \quad 2023, \\ \text{Test set:} & \quad 2024. \end{aligned} \quad (18)$$

This chronological partition is designed to evaluate the capability of the model to predict acoustic fields under future environmental conditions that are not included in the training period.

All statistics and acoustic priors that depend on the dataset, including normalization parameters and geo-steady fields, are calculated exclusively from the training set to prevent information leakage from the validation and test periods.

Summary of the Workflow

The complete sample-generation process can be summarized as

$$\begin{aligned}
&\{\text{temperature, salinity}\} \longrightarrow \text{3D sound-speed field,} \\
&\text{source location} \longrightarrow \text{50-level SSP,} \\
&\text{ETOPO2022} \longrightarrow \text{local bathymetry,} \\
&\{\text{SSP, bathymetry, source parameters}\} \longrightarrow \text{Bellhop3D environmental files,} \\
&\text{Bellhop3D} \longrightarrow \text{.shd pressure field,} \\
&\text{pressure field} \longrightarrow \text{3D TL field,} \\
&\{\text{source location, SSP}\} \longrightarrow \text{deep-learning input,} \\
&\text{3D TL field} \longrightarrow \text{deep-learning target.}
\end{aligned} \tag{19}$$

In this way, each Bellhop3D simulation establishes a one-to-one correspondence between the source/environmental conditions and the resulting 3D acoustic transmission-loss field. Repeating this procedure over different source locations, months, and years produces the large-scale dataset used for CMSM-Net training and evaluation.